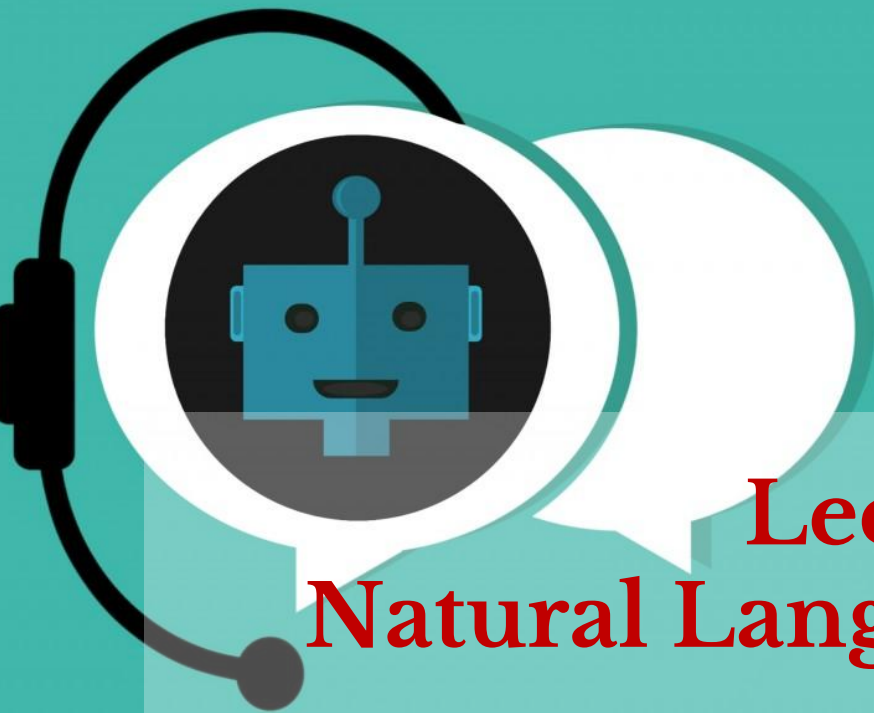




Lecture 1: Natural Language Processing

Instructor: Jackie CK Cheung & David I. Adelani

COMP-550

Winter 2026

J&M Chapter 1

About Jackie

Associate Professor at McGill

2021 -

- Associate Scientific Co-Director at Mila

Assistant Professor at McGill

2015 – 2021

PhD in Computer Science (Toronto)

2014

Research topics in my lab

- Natural language generation
- Automatic summarization
- Computational semantics
- Computational pragmatics
- Applications of NLP

About David

Assistant Professor at McGill

2024 -

- Core Academic Member at Mila

Senior Research Fellow at UCL

2022 – 2024

PhD in Computer Science (Saarland)

2023

Research topics in my lab

- Multilingual Natural language processing
- Machine translation
- Representation learning
- Speech processing

Preliminaries

Instructor: Jackie Chi Kit Cheung & David I. Adelani

Time and Loc.: 14:35 – 15:55 Leacock 26

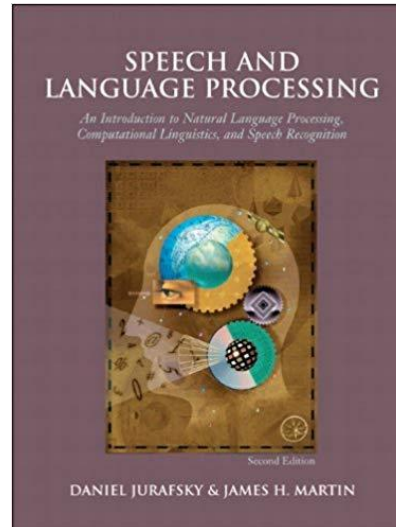
Office hours: TBD MC 108N (Jackie)
TBD (David)

TAs: Cléa Chataigner, Eric Huang, Soumya
Sharma, Pingsheng Li, Yiwei Zhang, Yijie
Zhang, Khaoula Chechbouni

Evaluation: 2 programming assignments (20%)
4 reading assignments (20%)
1 midterm (25%)
1 group project (35%)

Textbook

Jurafsky and Martin. *Speech and Language Processing*
(2nd edition)



Hard copy available at bookstore

Draft chapters of 3rd edition available online:

<https://web.stanford.edu/~jurafsky/slp3/>

Assignments

Two programming assignments (10% each x 2 = 20%)

Hand in online through myCourses

Programming to be done in Python 3.

Four reading assignments (5% each x 4 = 20%)

Covers advanced material and applications

Midterm

Worth 25% of your final grade

Time: Second half of March

Exact date and time to be confirmed based on room availability

More details as we approach the midterm date.

Final Project

Worth 35%.

Experiment on some language data set

Summarize and review relevant papers

Report on experiments

Must be done in teams of three

Coming up with a project idea:

- Extend a model we see in class
- Work on a relevant topic of interest
- Consult a list of suggested projects, to be posted

Project Steps

Paper or project proposal

Progress update

Final submission

Due dates to be announced

General Policies

Lateness policy for assignments:

- Grace period of 24 hours
- > 24 hours: accepted if it is convenient for us at our discretion

Plagiarism: just don't do it—I regularly catch and submit cases.

Language policy: In accord with McGill policy, you have the right to write essays and examinations in English or in French.

Generative AI Usage

Fine to use in an assistive manner

- Help understand course content
- Search for information
- Brainstorm ideas
- Edit writing

Must acknowledge use of this technology.

Not okay to use as primary means to complete tasks

- Feed in assignment questions to generate solutions
- Generate project report from scratch on a topic

Platforms

ed

Adopted by many CS courses

You'll be added soon

Most releases will be done via this platform

myCourses

Assignment and project submissions

Midterm

Grade release

Computational Linguistics and Natural Language Processing

LLMs – Research to Deployment

Timeline of (large) language model research

- 1990s to 2000s: statistical language models
- 2000s to 2010s: advances in deep learning
- 2018: ELMO – pretrained contextualized word embeddings
- 2019 – 2022: first Transformer-based large language models
- 2022 – now: ChatGPT and many subsequent models

Research to development in a few short decades!

What do you use LLMs for?

How Do Language Models Work?

Key insight: learn correlations between words in context

Language modelling:

Mary had a little _____

- *lamb* GOOD
- *accident* GOOD?
- *very* BAD
- *up* BAD

Do this at internet-scale with sophisticated statistical techniques (deep learning)!

What This Course Is About

- How did we get to large language models dominating NLP research?
- What was the progression of the field of NLP? Why did people try the methods that they did?
- What are some common tasks and paradigms involving natural language?
- How do we evaluate and analyze NLP systems?
- How are properties of natural language reflected in NLP research?

What This Course Is Not About

- The latest techniques in language modelling
- Deep learning / machine learning as a primary focus
 - We will touch on this, and you can do a final project that uses ML, but it is **not** the primary focus of the course.

Language is Everywhere

NEW | Hiker Julien Landry rescued days after fleeing up a tree to avoid bear

Hiker climbed a tree after a mother bear charged him - with incredible unexpected consequences

CBC News - Posted: Aug 21, 2014 12:30 PM PT | 3 | Last Updated: Aug 21, 2014 12:31 PM PT



Guest: hiker Julien Landry, 25, is told to be alone after he climbed a tree to escape a mother bear in Trout Creek, B.C. (Facebook)

4 shares



A Quebec man is in a stable condition in a Kelowna hospital after spending several days injured and alone in the forest after a mother bear attack.

After a day's work in the orchards around near 5 B.C., Julien Landry, 25, of Trois-Rivières, Que., was in the Trout Creek canyon when a bear charged, for it is not clear whether the bear and her cubs were bears but as they circled the tree below, Landry was in the branches for hours, growing increasingly ill.

"Eventually he fell asleep because he'd been working in the orchards," said RCMP Const. Jacques Lefebvre. "When he fell asleep he fell down off the tree and landed on some rocks in the creek."

Lying unconscious in the creek, it was a day and a half before Landry awoke. He eventually managed to drag himself out of the water but was too weak to walk.

A search and rescue team including an RCMP helicopter and a plane could not find him.

It was three more days before another hiker found Landry, who was unable to move. He had buried himself in dirt to keep warm. Landry suffered a concussion, bleeding in and broken vertebrae and was rushed to undergo emergency surgery. Doctors are in good recovery.

"I don't think he could have gotten himself lucky," said Lefebvre.

Scientists have some surprising news about going in the ocean



Orphaned bear cub was rescued in June after he hibernated alone



18.
Shall I compare thee to a Summers day?
Thou art more lovely and more temperate:
Rough winds do shake the darling buds of Maie,
And Sommers lease hath all too short a date:
Sometime too hot the eye of heaven shines,
And often is his gold complexion dimm'd,
And every faire from faire some-time declines,
By chance, or natures changing course vntim'd:
But thy eternal Sommer shall not fade,
Nor loose possession of that faire thou ow'st,
Nor shall death brag thou wandr'st in his shade,
When in eternal lines to time thou grow'st,
So long as men can breathe or eyes can see,
So long lives this, and this gives life to thee,



Languages Are Diverse

6000+ languages in the world

language

langue

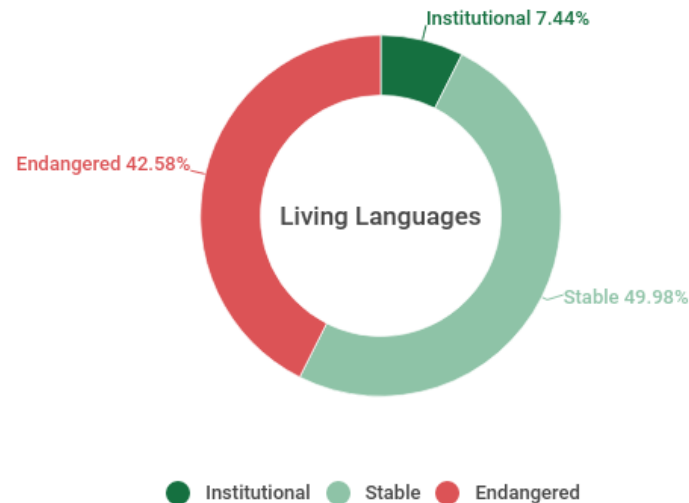
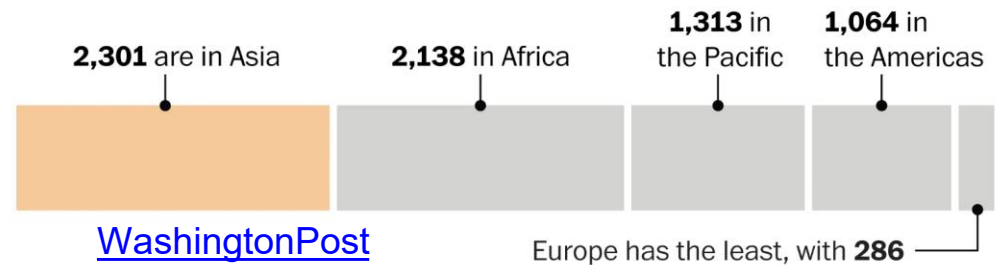
ভাষা

語言

idioma

Sprache

lingua



What is Language?

Some properties:

- Form of communication
- **Arbitrary** pairing between form and meaning
- Primarily vocal (exception: sign languages)
- Highly expressive and productive
- Nearly universal (barring developmental disorders)

How do these compare?

- Programming language (e.g., C, Python, Java)
- Vocalizations by your favourite animal
- Written English

Computational Linguistics (CL)

Modelling natural language with computational models and techniques

Domains of natural language

Acoustic signals, phonemes, words, syntax, semantics, ...

Speech vs. text

**Natural language understanding (or comprehension) vs.
natural language generation (or production)**

Computational Linguistics (CL)

Modelling natural language with computational models and techniques

Goals

Language technology applications

Scientific understanding of how language works

Computational Linguistics (CL)

Modelling natural language with computational models and techniques

Methodology and techniques

Gathering data: language resources

Evaluation

Statistical methods and machine learning

Rule-based methods

Natural Language Processing

Computational linguistics and **natural language processing (NLP)** are sometimes used interchangeably.

Slight difference in emphasis:

NLP

Goal: practical
technologies

Engineering

CL

Goal: how language
actually works

Science

Understanding and Generation

Natural language understanding (NLU)

Language to form usable by machines or humans

Natural language generation (NLG)

Traditionally, semantic formalism to text

More recently, also text to text

Most work in NLP used to be in NLU

c.f. linguistics, where most theories deal primarily with production

Personal Assistant App

Understanding

Call a taxi to take me to the airport in 30 minutes.

- Recognizing command, and type of command (what is a taxi? What does it mean to call it?)
- Figuring out time – time disambiguation
- The airport -> recognize it's a place; which airport?
- Location of pickup? Make implicit inference.

What is the weather forecast for tomorrow?

Generation

- Generate script if you need to interface with people
- Ask for clarifications to disambiguate
- Confirming for user

Machine Translation

I like natural language processing.



Automatische Sprachverarbeitung gefällt mir.

Understanding

- Syntactic parsing -> understand word order
- Meanings of words in source language
- Idiomatic expressions

Generation

- Generate according to grammar of target language
- Generate according to word meaning in target language
- Idiomatic expressions

Computational Linguistics

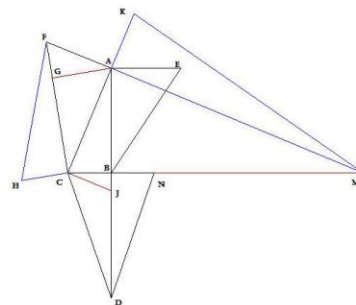
Besides new language technologies, there are other reasons to study CL and NLP as well.

The Nature of Language

First language acquisition

Chomsky proposed a **universal grammar**

Is language an “instinct”?



What innate knowledge must children already have in order to learn their mother tongue, given their exposure to linguistic inputs?

Train a model to find out!

The Nature of Language

Language processing

Some sentences are supposed to be grammatically correct, but are difficult to process.

Formal mathematical models to account for this.

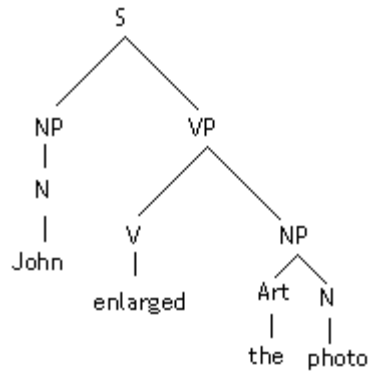
The rat escaped.

The rat the cat caught escaped.

*?? The rat the cat **the dog chased** caught escaped.*

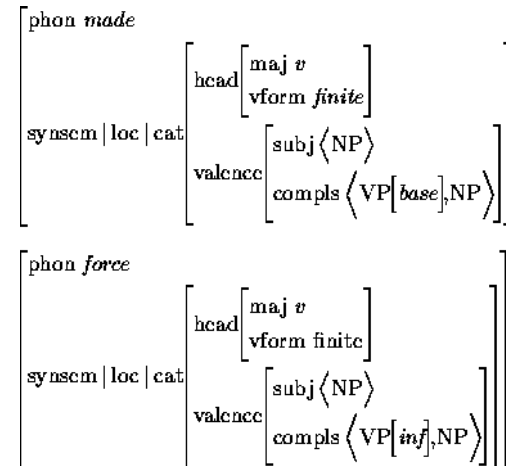
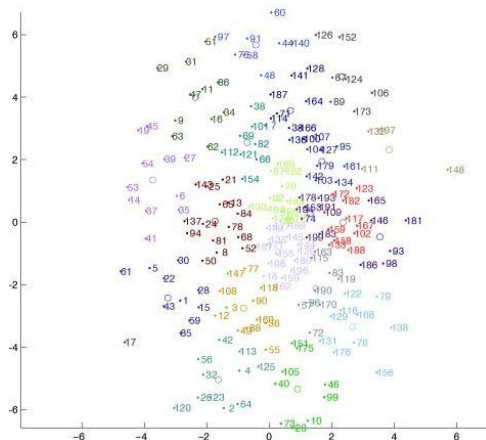
Mathematical Foundations of CL

We describe language with various formal systems.



cat + z > cats

cat + z	*SS	Agree	Max	Dep	Ident
catiz				*!	
catis				*!	*
catz		*!			
cat			*!		
cats					*



Mathematical Foundations of CL

Mathematical properties of formal systems and algorithms

Can they be efficiently learned from data?

Efficiently recovered from a sentence?

Complexity analysis

Implications for algorithm design

Types of Language

Text

In some sense, an idealization of spoken language.

Much of traditional NLP work has been on news text.

Clean, formal, standard English, but very limited!

Now, there is much work on other genres of text

Political texts, text messages, Twitter

Speech

Messier: disfluencies, non-standard language

Automatic speech recognition (ASR)

Text-to-speech generation

Domains of Language

The grammar of a language has traditionally been divided into multiple levels.

Phonetics

Phonology

Morphology

Syntax

Semantics

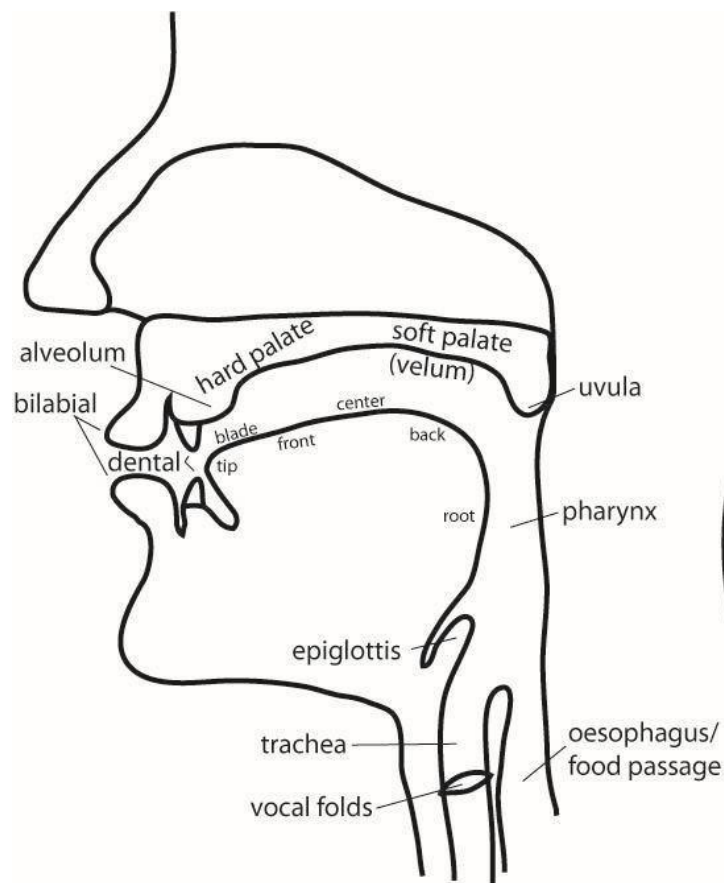
Pragmatics

Discourse

Phonetics

Study of the speech sounds that make up language

Articulation, transmission, perception



peach

[phi:tsh]

Involves closing of the lips, building up of pressure in the oral cavity, release with aspiration, ...

Vowel can be described by its formants, ...

Phonology

Study of the rules that govern sound patterns and how they are organized

<i>peach</i>	[phi:tsh]	/pi:ɸ/
<i>speech</i>	[spi:tsh]	/spi:ɸ/
<i>beach</i>	[bi:tsh]	/bi:ɸ/

The p in peach and speech are the same phoneme, but they actually are phonetically distinct!

Morphology

Word formation and meaning

antidisestablishmentarianism

anti- dis- establish -ment -arian -ism

establish

*establish**ment***

*establishment**arian***

*establishmentarian**ism***

***dis**establishmentarianism*

***anti**disestablishmentarianism*

Syntax

Study of the structure of language

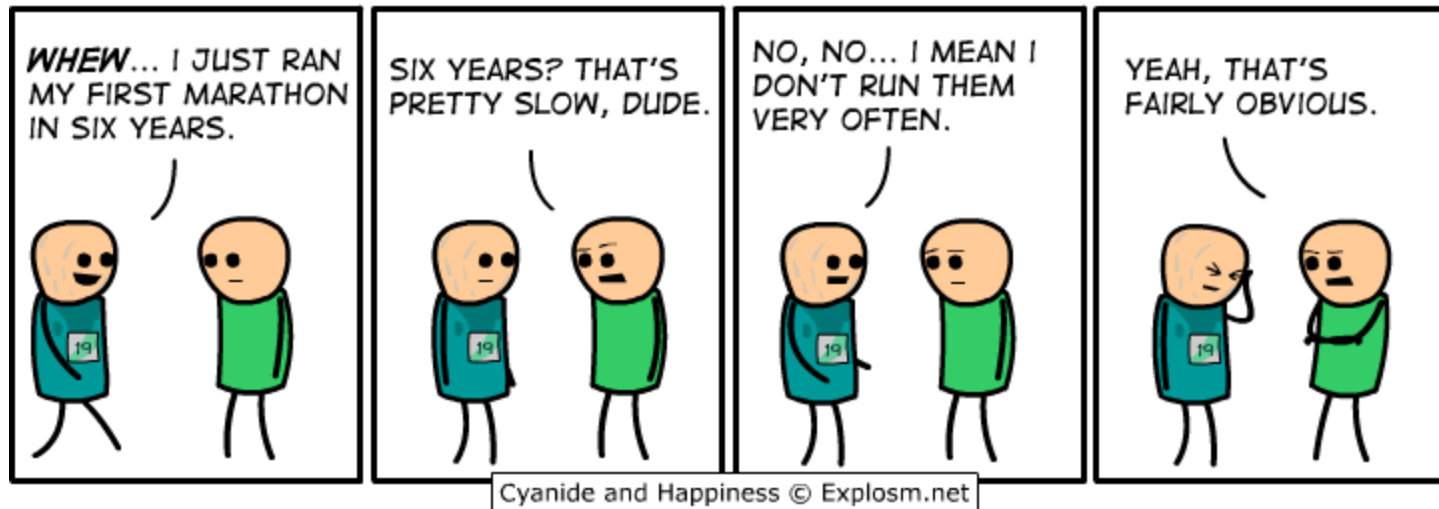
**I a woman saw park in the.*

I saw a woman in the park.

The first sentence is not well formed (it is **ungrammatical**), while the second one is.

- Words must be arranged in a certain order in a certain way to be a valid English sentence!

Syntax



<http://explosm.net/comics/1682/>

There are two meanings for the first sentence in the comic! What are they? This is called **ambiguity**.

Semantics

Study of the meaning of language

bank

Ambiguity in the **sense** of the word



Semantics

Ross wants to marry a Swedish woman.

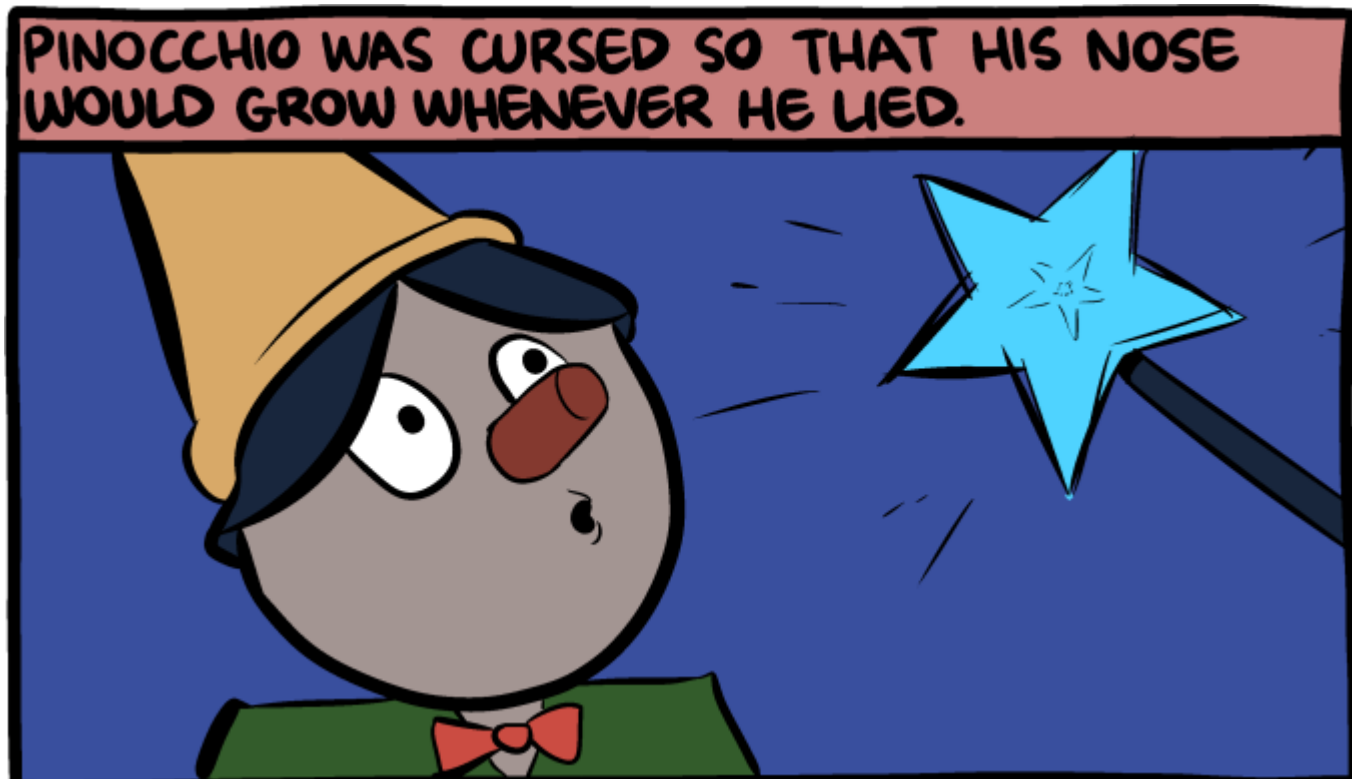


Pragmatics

Study of the meaning of language in context.

❑ Literal meaning (semantics) vs. meaning in context:

<http://www.smbc-comics.com/index.php?id=3730>



Pragmatics



Pragmatics



Pragmatics



Pragmatics – Deixis

Interpretation of expressions can depend on **extralinguistic** context

e.g., pronouns

_I think cilantro tastes great!

The entity referred to by *I* depends on who is saying this sentence.

Discourse

Study of the structure of larger spans of language (i.e., beyond individual clauses or sentences)

I am angry at her.

She lost my cell phone.

I am angry at her.

The rabbit jumped and ate two carrots.

NLP – the Technological Perspective

A combination of **pre-specified knowledge** and **machine learning from data**



Problem specification
Machine learning algorithms
Human annotations
Linguistic knowledge

...



Websites
News articles
Discussions
Knowledge bases

...

NLP Tools and Techniques

Major paradigms for NLP, not mutually exclusive:

Rule-based systems

- Often hand-engineered knowledge about language
- E.g., *heureux* -> *happy*

Machine learning

- Model learns about language through examples
- **Classification**: e.g., is this e-mail spam?
- **Sequence models**: make series of decisions
- Many other paradigms

Knowledge representation

- Formal structure to encode what model knows
- Logic? A large set of continuous-valued numbers?

Topics in COMP-550

Organized roughly by level of linguistic analysis and a corresponding technical approach (ML or otherwise)

NLP Topic	Linguistic layer	Techniques
Text classification	Words	Classification
Language modelling, POS tagging	Words (esp. syntactic structure of words)	Sequence models
Syntactic parsing	Syntactic structure	Structure prediction, dynamic programming
Computational semantics, coreference resolution	Meaning (semantics, discourse)	Logic, semi-supervised learning, neural models
Applications: MT, summarization, etc.	Various	Various

Applications in COMP-550

Last three weeks of the course focus on language technology applications and advanced topics

Possible topics:

- Cross-lingual NLP

- Vision and language

- Automatic summarization

- Machine translation

- Evaluation issues in NLP

- NLP and ethics

Accompanied by reading assignments!

Course Objectives

Understand the broad topics, applications and common terminology in the field

Prepare you for research or employment in CL/NLP

- Learn some basic linguistics

- Learn the basic algorithms

- Be able to read an NLP paper

Understand the challenges in CL/NLP

- Answer questions like “Is it easy or hard to...”